

Lucas Y. Tian, Ph.D.

NINDS K99 Postdoctoral Associate
Laboratory of Neural Systems
The Rockefeller University
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EDUCATION

2012 - 2019 **Ph.D.** in Neuroscience, UC San Francisco, CA (Advisor: Michael Brainard)
2007 - 2011 **B.A.** with Honors in Biology, specializing in Neuroscience, University of Chicago, IL

RESEARCH

2019 - Present **Postdoctoral Scholar**, The Rockefeller University, New York, NY
 Advisor: Winrich Freiwald; Co-advisors: Josh Tenenbaum, Xiao-Jing Wang
 Topic: Neural substrates of compositional generalization, in a macaque drawing task.

2019 (May-Dec) **Visiting Postdoctoral Scholar**, MIT, Cambridge, MA
 Advisor: Josh Tenenbaum; Co-advisors: Winrich Freiwald, Xiao-Jing Wang
 Topic: Cognitive computational modeling of human drawing.

2012 - 2019 **Graduate Student**, UCSF, San Francisco, CA
 Advisor: Michael Brainard
 Thesis: Neural mechanisms of motor skill flexibility in songbirds ([link](#))

AWARDS AND HONORS

General

2025 MSN Extramural Postdoctoral Invited Speaker, Mt. Sinai
2025 SPINES Extramural Postdoctoral Invited Speaker, NYU
2025 NEUROYES Extramural Postdoctoral Invited Speaker, University of Rochester
2023 NINDS K99/R00 Transition to Independence Award
2020 NIH BRAIN F32 NRSA Postdoctoral Fellowship
2014 Herbert W. Boyer PIBS Fellowship, UCSF

Travel grants

2026 Neural Control of Movement Scholarship
2025 Postdoctoral Association Career Development Award, Rockefeller
2025 COSYNE Presenters Travel Grant
2016 Graduate Division Travel Award, UCSF

PEER-REVIEWED PUBLICATIONS

Tian LY, Garzón Gupta K, Hanuska DJ, Rouse AG, Eldridge MA, Schieber MH, Wang XJ, Tenenbaum JB, Freiwald WA (2025). Neural representation of action symbols in primate frontal cortex. *Nature* (accepted in principle).

Preprint: [bioRxiv 2025.03.03.641276](https://doi.org/10.1101/2025.03.03.641276). Presentations: SfN 2024 poster ([link](#)), Cosyne 2025 talk ([link](#)).

Tian LY, Warren TL, Mehaffey WH, Brainard MS (2023). Dynamic top-down biasing implements rapid adaptive changes to individual movements. *eLife* 12:e83223. [Link](#)

Veit L, **Tian LY**, Monroy Hernandez CJ, Brainard MS (2021). Songbirds can learn flexible contextual control over syllable sequencing. *eLife* 10: e61610. [Link](#)

Tian LY, Ellis K, Kryven M, Tenenbaum JB (2020). Learning abstract structure for drawing by efficient motor program induction. *Advances in Neural Information Processing Systems*, 33 [oral presentation]. [Link](#)

Tian LY, Brainard MS (2017). Discrete circuits support generalized versus context-specific vocal learning in the songbird. *Neuron* 96, 1-10. [Link](#), [Commentary](#), [News](#)

PREPRINTS

Tian LY, Hanuska DJ, Garzón Gupta K, Liu Y, Wang XJ, Tenenbaum JB, Freiwald WA (2026). Neural representation of action grammar structure in primate frontal cortex. [bioRxiv 2026.01.18.700235](#)

FUNDING

2023 - Present	NINDS K99/R00 Transition to Independence Award Title: Investigating Symbolic Computation in the Brain: Neural Mechanisms of Compositionality (5K99NS131585) (PI: Tian, LY) Total Award Amount (including Indirect Costs): \$268,542 (2-year K99 phase), \$747,000 (3-year R00 phase, estimated)
2021 - Present	Simons Foundation SCGB Pilot Award (key personnel – contributing co-author) Title: Investigating Symbol-like Reasoning in the Brain: Neural Mechanisms of Compositional Action Planning (876120SPI) (PI: Freiwald, WA) Total Award Amount (including Indirect Costs): \$770,000
2020 - 2023	NIH BRAIN F32 NRSA Postdoctoral Fellowship Title: The planning of new compositional action sequences guided by interpretation of ambiguous sensory data in a novel drawing task (F32MH125573) (PI: Tian, LY) Total Award Amount: \$209,778

INVITED TALKS

Competitively selected conference talks

2026	Neural Control of Movement Team Oral (Panel), Kobe, Japan
2025	Society for Neuroscience Nanosymposium, San Diego, USA
2025	COSYNE Main Meeting (3% of submissions), Montreal, Canada (Recording)
2024	Ascona Circuits Meeting, Ascona, Switzerland (Slides)
2020	NeurIPS (1% of submissions) (Recording)

Symposia

2025	Kavli Internal Symposium, Rockefeller, New York, USA
2025	Lipschultz Symposium, Icahn School of Medicine (Mt. Sinai), New York, USA
2025	The Social Brain Symposium, Rockefeller and Columbia, New York, USA
2021	Object Cognition Workshop, Yale, New Haven, USA (virtual)
2018	Birds Bats Brains Meeting, UC Berkeley, USA

Departmental talks

2026 Departments of Neuroscience and Computer Science, UT Austin, USA
 2026 NEUROYES Extramural Postdoctoral Invited Speaker, University of Rochester, USA
 2025 Enigma Project, Stanford, USA (virtual)
 2025 MSN Extramural Postdoctoral Invited Speaker, Mt. Sinai, USA
 2025 Department of Neurobiology, University of Pittsburgh, USA
 2025 German Primate Center, Leibniz Institute for Primate Research, Göttingen, Germany
 2023 Kavli Internal Seminar series, Rockefeller, New York, USA
 2021 Center for Brains, Minds & Machines, MIT, Cambridge, USA ([Link to recording](#))
 2021 PDA Summer Seminar Series, Rockefeller, New York, USA
 2016 Asilomar Neuroscience Retreat, UC San Francisco, USA

Lab talks

2025 UCSD, Bria Long
 2025 Western University, Jörn Diedrichsen, Paul Gribble, and Andrew Pruszynski (virtual)
 2025 Netherlands Institute for Neuroscience, Pieter Roelfsema
 2025 Princeton, Tim Buschman
 2025 NYU, Marcelo Mattar
 2025 MIT, Ev Fedorenko
 2025 Newcastle University, Stuart Baker (virtual)
 2025 NYU, Michael Long
 2022 UCSF, Michael Brainard
 2021 Yale, Ilker Yildirim (virtual)
 2020 NYU, Brenden Lake and Todd Gureckis (virtual)
 2020 MIT, Laura Schulz (virtual)
 2016 UCSF, John Houde

PROFESSIONAL SERVICE

Workshop organization

2026 Co-organizer (w/ Mathias Sablé-Meyer), COSYNE Workshop, “Symbols as foundational to the biological basis of intelligent behavior”

Other service

2023 - Present Peer-reviewed for: eLife, Cognitive Science Society, Cognitive Computational Neuroscience
 2014 - 2019 Co-organizer, Systems Neuroscience Research in Progress Seminar Series, UCSF
 2013 - 2019 Co-organizer, Special Topics Seminar Series, UCSF

SELECT CONFERENCE ABSTRACTS

2025 Cognitive Computational Neuroscience, Amsterdam, The Netherlands
 2025 Society for the Neural Control of Movement, Panama City, Panama
 2024 Society for Neuroscience, Chicago, IL ([Link to poster](#))
 2024 Ascona Circuits Meeting, Ascona, Switzerland

MENTORSHIP

Research assistants

2024 - Present Daniel Hanuska, research assistant, Rockefeller

2022 - 2024 Kedar Garzón Gupta, research assistant, Rockefeller
(*Current position: Ph.D. Student at Columbia*)

Ph.D. rotation students

2024 Siddhartha Sharma, Freiwald lab, Rockefeller
2021 Yanis Tazi, Freiwald lab, Rockefeller
2018 Eszter Kish, Brainard lab, UCSF
2014 Rachel Care, Brainard lab, UCSF

Interns and other mentorship

2026 - Present Kateryna Panikhina, Bard-Rockefeller intern, Rockefeller
2024 - 2025 Valerie Calligy, Shenoy Undergraduate Research Fellow (SURFiN), Simons Foundation, Rockefeller
2024 - 2025 Xuan Ma, Bard-Rockefeller intern, Rockefeller
2024 Daniel Dolnik, intern, Cornell Tech Master's Program
2024 Thea Wu, summer intern (Summer Undergraduate Research Fellowship, SURF), Rockefeller
2022 Alejandra Urquieta, summer intern, SURF, Rockefeller
2021 Sam Coolsaet, medical intern, Rockefeller
2021 Alex Gu, summer intern, SURF, MIT
2017 Christian Jose Monroy Hernandez, HHMI Exceptional Research Opportunities Program (EXROP), UCSF (co-mentor: Lena Veit)

OUTREACH

2024 - 2025 Mentor, Simons Foundation SURFiN, Rockefeller
Mentee: Valerie Calligy
2021 Mentor, Neuromatch Academy (virtual)
Mentees: Pulido Arias, Lourdes Baztan Bitopoulos, Joseph Gonzalez, May Xia, Mark Zarutin
2017 Mentor, HHMI EXROP, UCSF
Mentee: Christian Jose Monroy Hernandez (co-mentor: Lena Veit)
2014 Judge, Synopsys Science and Technology Championship, San Jose, CA
2013 Invited speaker, PITCH, Center for Education Partnerships (CEP), UCSF
2013 - 2014 Mentor, Fairposium at Burton High School, CEP, UCSF

TEACHING

2014 TA, BMS117: Infection & Host Response (Neurophysiology unit), UCSF
2012 - 2013 Scientist-Teacher, Raul Wallenberg High School, Science and Health Education Partnership, UCSF
2012 English Teacher, San Gabriel Kindergarten, Los Nogales, Peru
2010 Teacher, Neuroscience of Illusions, Cascade!, University of Chicago

ADVANCED COURSES

2019 Methods in Computational Neuroscience, Marine Biological Laboratory, MA
2018 Jr. Workshop on Mechanistic Cognitive Neuroscience, Janelia Research Campus, VA
2018 Brains, Minds and Machines, Marine Biological Laboratory, MA

2017 Mining and Modeling of Neuroscience Data, Redwood Center, UC Berkeley, CA

REFERENCES

Winrich Freiwald, Ph.D.

Denise A. and Eugene W. Chinery Professor
Laboratory of Neural Systems, Rockefeller University
1230 York Avenue, New York, NY 10065
wfreiwald@rockefeller.edu
(212) 327-8000
Relationship: Postdoctoral advisor

Michael Brainard, Ph.D.

Professor of Physiology and Psychiatry
Departments of Physiology and Psychiatry, UCSF/HHMI
675 Nelson Rising Lane, Room 514E, San Francisco, CA 94158
michael.brainard@ucsf.edu
(415) 502-7344
Relationship: Ph.D. advisor

Joshua Tenenbaum, Ph.D.

Professor of Computational Cognitive Science
Department of Brain and Cognitive Sciences, MIT
Building 46-4015, 77 Massachusetts Avenue, Cambridge, MA 02139
jbt@mit.edu
(617) 452-2010
Relationship: Postdoctoral co-advisor

Xiao-Jing Wang, Ph.D.

Professor of Neural Science
Center for Neural Science, NYU
4 Washington Place, Room 752, New York, NY 10003
xjwang@nyu.edu
(212) 998-3677
Relationship: Postdoctoral co-advisor